

279V+ **Smart** Battery Cabinet



279V+

NOMINAL
VOLTAGE

50kWh

NAMEPLATE
ENERGY

25kW

MAX RATED
POWER

6K+

CYCLE LIFE
@ 80% DOD

-40/+55°C

OPERATING
TEMP

IP54

ENCLOSURE
RATING

20yr

DESIGN
LIFE

Smart

ENERGIOS™
ENABLED

Series B 279V+ Smart Battery

- Sodium-Ion NFPP
- Air-Cooled Cabinet
- EnergiOS™ Smart EMS
- IP54 Outdoor-Rated



Built for critical infrastructure resiliency and demanding commercial & industrial use cases, the Series B 279V+ Smart Battery Cabinet prioritizes compliance, field-validated performance, and bankable warranties — delivering a **20-year design life** and **6,000+ cycles at 80% DoD** with zero maintenance and no HVAC required.

INCLUDES AN OUTDOOR, IP54-RATED ENCLOSURE CONTAINING:

- | | |
|-----------------------------------|------------------------------------|
| • Sodium-Ion NFPP battery modules | • Battery Management System (BMS) |
| • Air-cooled thermal management | • EnergiOS™ Smart EMS platform |
| • Auxiliary safety systems | • Fire suppression & gas detection |

KEY BENEFITS



Safety-First Design

Sodium-Ion NFPP chemistry eliminates thermal runaway risk — enabling safer siting, simpler AHJ approvals, and no fire suppression complexity.

- ✓ Zero thermal runaway incidents
- ✓ No toxic gas emissions
- ✓ Air-shippable — faster deployment
- ✓ Cleaner AHJ approvals



Extreme Performance

Operates from -40°C to +55°C without derating or HVAC infrastructure — ideal for outdoor installations, cold storage, and industrial environments.

- ✓ -40°C to +55°C operation
- ✓ No HVAC required
- ✓ Minimal capacity derating
- ✓ Fast charging capability



Intelligent & Bankable

Built-in EnergiOS™ Smart EMS transforms the cabinet into an active energy asset — optimizing dispatch, enabling VPP participation, and maximizing ROI.

- ✓ 6,000+ cycles @ 80% DoD
- ✓ 20-year design life
- ✓ ITC eligible with solar PV
- ✓ Demand response & VPP ready

WHERE THE SERIES B PERFORMS



MANUFACTURING / COLD STORAGE / DISTRIBUTION

Peak Demand Management

Reduce demand charges 30–50% at manufacturing facilities, cold storage, and distribution centers while maintaining full capacity in extreme temperatures.



DATA CENTERS / HEALTHCARE / TELECOM

Critical Infrastructure

Guaranteed uptime for data centers, hospitals, and telecom — with air-cooled simplicity, intelligent monitoring, and no fire suppression complexity.



FORKLIFT / WAREHOUSE / PORT / AIRPORT

Material Handling & Fleet

Fast charging and extreme temp tolerance for 24/7 operations — forklift fleets, warehouse automation, port equipment, and airport ground support.



SOLAR / WIND / CAMPUS / COMMUNITY

Renewable Integration

Long-duration storage with 20-year design life for bankable solar + storage projects, campus microgrids, wind smoothing, and community solar.

THREE TECHNOLOGIES. ONE INTELLIGENT PLATFORM.

Not Just a Cabinet — An Intelligent Energy Platform

The Series B integrates with Coulomb Technology's Smart EMS platform — a suite of three proprietary software technologies that transform your battery cabinet into an active, revenue-generating energy asset. Monitor, optimize, and control your energy in real time from any device.



The intelligent operating system for your energy storage. Provides real-time monitoring, automated charge/discharge scheduling, and seamless solar integration.

- Real-time SOC, power flow & health monitoring
- Automated TOU & solar charge scheduling
- Remote access via web & mobile app
- Alerts, diagnostics & OTA firmware updates



Machine learning algorithms that learn energy usage patterns, weather forecasts, and utility rates to autonomously optimize dispatch and maximize savings.

- Predictive load forecasting & solar yield modeling
- Autonomous dispatch optimization
- Dynamic rate arbitrage (TOU, demand charges)
- Continuous learning — improves over time



Advanced grid interaction capabilities enabling virtual power plant participation, demand response programs, and utility incentive monetization.

- Virtual power plant (VPP) participation
- Demand response & utility incentive programs
- Grid frequency regulation support
- Export control & islanding management

8 INTELLIGENT BMS PROTECTIONS

Overcharge Protection	Overcurrent Protection	Overdischarge Protection	Overheat Protection
Temperature Management	Short Circuit Protection	Balanced Power	Reverse Connection

TOTAL COST OF OWNERSHIP ADVANTAGE

20_{yr} DESIGN LIFE VS. 3-5 YR LEAD-ACID	Zero MAINTENANCE COST OVER LIFETIME	6K+ CYCLES @ 80% DOD VS. 300-500 LEAD-ACID	50% DEMAND CHARGE REDUCTION POTENTIAL
---	--	---	--

- **No replacement costs** — 20-year design life eliminates the 3-4 lead-acid replacement cycles over the same period.
- **Zero maintenance overhead** — no watering, no equalization, no acid spill cleanup, no HVAC servicing.
- **ITC eligible** — qualifies for 30% federal Investment Tax Credit when paired with solar PV systems.
- **Utility incentives** — GridIQ™ enables demand response and VPP revenue participation for additional income streams.

RESEARCH & DEVELOPMENT PARTNERS



SERIES B — 279V+
CABINET



NOMINAL VOLTAGE

279V+

High-voltage DC bus

NAMEPLATE ENERGY

50kWh

42.5 kWh usable at 85% DoD

CYCLE LIFE

6K+

At 80% DoD, 25°C

MAX RATED POWER

25kW

Continuous discharge

OPERATING TEMP

−40/+55°C

No HVAC — any climate

ENCLOSURE RATING

IP54

Outdoor-rated direct install

PARAMETER	VALUE / DESCRIPTION	UNIT
NOMINAL VOLTAGE	279 (high-voltage DC bus)	VDC
NAMEPLATE ENERGY CAPACITY	50	kWh
USABLE ENERGY (85% DOD)	42.5	kWh
MAX RATED POWER	25	kW
CYCLE LIFE	6,000+ at 80% DoD, 25°C	—
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
OPERATING TEMP (DISCHARGE)	−40°C to +55°C	—
CHARGING TEMP RANGE	−10°C to +45°C	—
DESIGN LIFE	20 years	—
ENCLOSURE (IP RATING)	Outdoor Cabinet — IP54	—
THERMAL MANAGEMENT	Air-Cooled (No HVAC Required)	—
BMS COMMUNICATION	Integrated BMS, Modbus TCP / CAN Bus	—
SMART EMS	EnergiOS™ — Real-time monitoring, TOU scheduling, OTA updates	—
AUXILIARY SYSTEMS	Fire Suppression, Gas Detection, Explosion Venting	—

CERTIFICATIONS

UL 1973

UL 9540

UN 38.3

IEC 62619

Cell & Module Level

UL 9540A TESTING

Cell: Tested & Passed

Module/System: In Progress

WARRANTY

Cell: 3 Years

System: 5 Years

Extended warranty available

FACTORY ACCEPTANCE TEST (FAT) SCOPE

- ✓ Visual, Mechanical & Torque Inspection

✓ 3 × 0.125C Capacity/Energy Verification (85% DOD)
- ✓ Protections & Isolation Tests

✓ Baseline OCV Measurement
- ✓ Functional & Conditioning Test

✓ 10 × 0.5C Stability Cycles with Power Checks